

MLF Week 4 : Linear Algebra - Eigenvalues and eigenvectors

Lecture 1: Linear and Polynomial Regression

Linear Regression

Given data $\{(x_1, y_1), \dots, (x_n, y_n)\}$, $x_i \in \mathbb{R}^d$, $y_i \in \mathbb{R}$, $i=1 \dots n$

$$L(\theta) = \frac{1}{2} \sum_{i=1}^n (x_i^T \theta - y_i)^2$$

Minimize L : Define $A = \begin{bmatrix} x_1^T \\ x_2^T \\ \vdots \\ x_n^T \end{bmatrix} \leftarrow \text{Feature matrix}, Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$

$$A\theta = \begin{bmatrix} x_1^T \theta \\ \vdots \\ x_n^T \theta \end{bmatrix} \quad A\theta - Y = \begin{bmatrix} x_1^T \theta - y_1 \\ \vdots \\ x_n^T \theta - y_n \end{bmatrix} \quad (A\theta - Y)^T (A\theta - Y) = \sum_{i=1}^n (x_i^T \theta - y_i)^2$$

So, $L(\theta) = \frac{1}{2} (A\theta - Y)^T (A\theta - Y)$

Minimizing L : $\nabla_{\theta} L(\theta) = 0$

$$\nabla_{\theta} ((A\theta - Y)^T (A\theta - Y)) = 0$$

$$\Leftrightarrow A^T (A\theta - Y) = 0$$

$$\Leftrightarrow \boxed{(A^T A) \theta = A^T Y} \rightarrow \text{Least square solution}$$

Note! Can we write $\theta = (A^T A)^{-1} A^T Y$?

Yes, if A is full rank [why? full rank $A \Rightarrow A^T A$ is invertible.

How to argue this? Show that $\text{rank}(A) = \text{rank}(A^T A)$

by showing $N(A) = N(A^T A)$.

H.w.

[Take an $x \in N(A)$ & show $x \in N(A^T A)$ and the other way]

Maximum likelihood and least squares:

Suppose the data is generated according to the following model:

$$x \rightarrow \boxed{\text{Linear Model}} \rightarrow y = \theta^T x + \epsilon$$

ϵ zero-mean, e.g., $\epsilon \sim \text{Gaussian with mean 0 and variance } (\frac{1}{\beta})$

Dataset $\mathcal{D} = \{(x_i, y_i), i=1 \dots n\}$ generated in an i.i.d fashion
 $\downarrow$
independent & identically distributed

Maximum likelihood approach:

$$L(\theta) = \prod_{i=1}^n \frac{\sqrt{\beta}}{\sqrt{2\pi}} \exp\left(-\frac{\beta}{2} (y_i - \theta^T x_i)^2\right) \rightarrow \text{why this function? (Module 6 on probability)}$$

ML approach: Find a θ that maximizes $L(\theta)$

Instead of $\max_{\theta} L(\theta)$, we could consider $\max_{\theta} \log L(\theta)$

Since $\log$ is increasing, the optima of these two problems are the same.

$$\log L(\theta) = \frac{n}{2} \log \beta - \frac{n}{2} \log 2\pi - \beta \left[\frac{1}{2} \sum (y_i - \theta^T x_i)^2 \right]$$

$\uparrow$
is the least squares objective that we minimized w.r.t θ

Message: "maximizing $\log L(\theta)$ " is the same as "min $\frac{1}{2} \sum (y_i - \theta^T x_i)^2$ "

(or) Least squares regression solves a maximum likelihood estimation problem under a linear model.

Polynomial regression:

Consider one-dimensional data

$$\mathcal{D} = \{ (x_i, y_i), i=1 \dots n \} \quad x_i, y_i \in \mathbb{R}, \forall i$$

Previously, we tried fitting a line through this data.

Here, we generalize to any polynomial, say of degree m .

Transformed features:

$$\begin{aligned} \hat{y}(x) &= \theta_0 + \theta_1 x + \theta_2 x^2 + \dots + \theta_m x^m \\ &= \sum_{j=0}^m \theta_j \phi_j(x), \quad \text{where } \phi_j(x) = x^j \end{aligned}$$

For a given x , the transformed feature vector $\phi(x) = (1, x, x^2, \dots, x^m)$

$$\hat{y}(x) = \theta^T \phi(x), \quad \theta = (\theta_0, \dots, \theta_m)$$

Using these transformed features, perform linear regression.

$$A = \begin{bmatrix} \phi(x_1)^T \\ \vdots \\ \phi(x_n)^T \end{bmatrix}, \quad Y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$(A^T A) \theta = A^T Y$$

Different wrt regular least squares is that we are using transformed features & then performing regression.

Regularized version of linear regression (a.k.a.) Ridge regression

Instead of solving $\min_{\theta} \frac{1}{2} \sum_{i=1}^n (x_i^T \theta - y_i)^2$, we solve the following regularized version:

$$\min_{\theta} \left\{ \bar{J}(\theta) = \frac{1}{2} \sum_{i=1}^n (\hat{x}_i^T \theta - y_i)^2 + \underbrace{\lambda \|\theta\|^2}_{\text{regularization term}} \right\}$$

Repeat the calculation leading to least squares solution, we obtain

$$(A^T A + \lambda I) \theta_{\text{reg}} = A^T Y$$

$$\theta_{\text{reg}} = (A^T A + \lambda I)^{-1} A^T Y$$

H.W.: Show $(A^T A + \lambda I)$ is invertible even if A is not full rank.

Note: ^{regularization} Parameter λ controls overfitting.

Too small a $\lambda \rightarrow$ overfitting

Too large a $\lambda \rightarrow$ underfitting

QN : 3,4

Lecture 2 : Eigenvalues and Eigenvectors

Eigen values and eigenvectors

Motivation: Ordinary differential equations

$$(*) \rightarrow \frac{dv}{dt} = 4v - 5w, \quad v=8 \text{ at } t=0, \quad \frac{dw}{dt} = 2v - 3w, \quad w=5 \text{ at } t=0$$

$$u(t) = \begin{bmatrix} v(t) \\ w(t) \end{bmatrix} \quad u(0) = \begin{bmatrix} 8 \\ 5 \end{bmatrix}$$

(*) can be re-written as

$$\frac{du}{dt} = Au, \quad \text{where } A = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}$$

and $u = u(0)$ at $t=0$.

Simple case of a single equation: $\frac{dy}{dt} = ay$ — (**)

Solution: $u(t) = e^{at} u(0)$ ← easy to see that this is a solution of (**)

$$(**) \begin{cases} \text{is unstable if } a > 0 & (e^t \rightarrow \infty \text{ as } t \rightarrow \infty) \\ \text{is neutrally stable if } a = 0 \\ \text{is stable if } a < 0 \end{cases}$$

Case of a 2d-parameter:

Want solutions of the following form for the system (*):

$$v(t) = e^{\lambda t} y, \quad w(t) = e^{\lambda t} z$$

$$(\text{or}) \quad u(t) = e^{\lambda t} x, \quad \text{where } x = \begin{bmatrix} y \\ z \end{bmatrix}$$

If we substitute this in $\frac{du}{dt} = Au$, we obtain

$$\lambda e^{\lambda t} y = 4 e^{\lambda t} y - 5 e^{\lambda t} z$$

$$\lambda e^{\lambda t} z = 2 e^{\lambda t} y - 3 e^{\lambda t} z$$

Take out $e^{\lambda t}$, we get

$$\begin{cases} 4y - 5z = \lambda y \\ 2y - 3z = \lambda z \end{cases}$$

$(\Rightarrow)$

$$Ax = \lambda x$$

$\rightarrow$ eigenvalue equation

eigenvalue

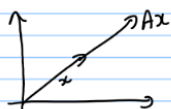
eigenvector
(non-trivial)

Message: Can solve $\frac{dy}{dt} = Au$ w/ solutions of the form $u(t) = e^{\lambda t} x$ if

$Ax = \lambda x$ can be solved.

Eigenvalues & eigenvectors:

For a matrix A , λ is an eigenvalue & $x \neq 0$ is an eigenvector if $Ax = \lambda x$.



If x is an eigenvector, then

A either stretches it or shrinks it, but A does not change the direction of x .

Case $\lambda = 0$: $Ax = 0 \Rightarrow x \in N(A)$

Examples:

① Projection matrix P

e.g. project onto a plane



(i) $Px = x$ for any x in the plane.

$\Rightarrow \lambda = 1$ is an eigenvalue & any x in the plane is an eigenvector

(ii) Consider an x that is perpendicular to the plane

$$Px = 0$$

So, $\lambda = 0$ is another eigenvalue & any $x \perp$ plane is an eigenvector.

② Permutation matrix $\rightarrow$ take the identity matrix & shuffle its rows.

e.g. $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

$$Bx = x \text{ for } x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad Bx = -x \text{ for } x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Find the eigenvalues:

$$Ax = \lambda x \Leftrightarrow (A - \lambda I)x = 0 \Leftrightarrow (A - \lambda I) \text{ is singular}$$

$$\Leftrightarrow \boxed{\det(A - \lambda I) = 0}$$

↓

characteristic polynomial of a matrix A .

If A is $n \times n$, then the characteristic polynomial is of degree n

$$(a_{11} - \lambda) \dots (a_{nn} - \lambda) = 0$$

" n " roots of characteristic polynomial = eigenvalues

for eigenvectors: Given λ , want $(A - \lambda I)x = 0$

$$\text{i.e., } x \in N(A - \lambda I)$$

↓
find this to obtain an eigenvector corresponding to eigenvalue λ
(e.g. use Gaussian elimination)

Example:

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\lambda^2 - 6\lambda + 8 = 0$$

$$\lambda_1 + \lambda_2 = 6, \quad \lambda_1 \lambda_2 = 8 \quad \left(\text{Check: } \lambda_1 + \lambda_2 = \text{trace}(A) = \text{sum of diagonal entries.} \right. \\ \left. \lambda_1 \lambda_2 = \det(A) = \text{product of eigenvalues} \right)$$

$$\text{Eigenvalues: } \lambda_1 = 4, \quad \lambda_2 = 2$$

$$A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$(A - 4I)x = 0 \text{ for } x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$(A - 2I) = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$(A - 2I)x = 0 \text{ for } x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

} The eigenvectors are linearly independent

Variation! Suppose

$$B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$B + 3I = A$$

$$Ax = (B+3I)x = \lambda x + 3x = (\lambda+3)x$$

H.W. Eigenvectors of B = eigenvectors of A .

Remarks:

- ① Suppose $Ax = \lambda_1 x$ and $Bx = \lambda_2 x$

Then, can we claim $(A+B)x \stackrel{?}{=} (\lambda_1 + \lambda_2)x$

NO: since the x 's need not be the same for A & B
 i.e., $Ax_1 = \lambda_1 x_1$ & $Bx_2 = \lambda_2 x_2$ with x_1 not necessarily equal to x_2 .

- ② Symmetric matrix has "real" eigenvalues.

A matrix whose eigenvalues are not real: $A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$

$$\det(A - \lambda I) = \lambda^2 + 1$$

$$\lambda_1 = i, \quad \lambda_2 = -i$$

- ③ A 2×2 matrix where we do not get "two independent" eigenvectors:

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix} \quad \lambda_1 = \lambda_2 = 3$$

$$(A - \lambda I) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ is a eigenvector \&}$$

there is no other eigenvector that is linearly independent of x .

QN : 4,5,6

Lecture 3 : Diagonalization of a matrix

Similarity and diagonalization

Def.: A matrix A is diagonalizable if $\exists$ an invertible matrix S such that

$$S^{-1}AS = \Lambda$$

↘ diagonal matrix

"Matrix being diagonalizable is tied to having enough independent eigenvectors"

Suppose A is a $n \times n$ matrix with n linearly independent eigenvectors, say $\{x_1, \dots, x_n\}$ with corresponding eigenvalues $\{\lambda_1, \dots, \lambda_n\}$

Then, A is diagonalizable. Why?

$$\text{Let } S = \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \dots & x_n \\ | & | & & | \end{bmatrix}$$

Is S invertible? Yes, because $\text{rank}(S) = n$ because $\{x_1, \dots, x_n\}$ is a linearly independent set

$$\begin{aligned} AS &= A \begin{bmatrix} | & & | \\ x_1 & \dots & x_n \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | \\ Ax_1 & \dots & Ax_n \\ | & & | \end{bmatrix} \\ &= \begin{bmatrix} | & & | \\ \lambda_1 x_1 & \dots & \lambda_n x_n \\ | & & | \end{bmatrix} = \begin{bmatrix} | & & | \\ x_1 & \dots & x_n \\ | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \\ &= S \Lambda, \text{ where } \Lambda = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \end{aligned}$$

$$\text{So, } AS = S \Lambda \Leftrightarrow S^{-1}AS = \Lambda$$

Claim: If λ_1, λ_2 are the eigenvalues with corresponding eigenvectors x_1, x_2 , and $\lambda_1 \neq \lambda_2$,

Then, $\{x_1, x_2\}$ is a linearly independent set.

Pr:

Suppose $c_1 x_1 + c_2 x_2 = 0$ — ①

$$A(c_1 x_1 + c_2 x_2) = 0$$

$$c_1 \lambda_1 x_1 + c_2 \lambda_2 x_2 = 0 \quad \text{--- ②}$$

$$c_1 (\lambda_1 - \lambda_2) x_1 = 0 \quad \leftarrow \text{②} - \lambda_2 \text{①}$$

$$\lambda_1 \neq \lambda_2, x_1 \neq 0 \Rightarrow c_1 = 0$$

Similarly, $c_2 = 0$

$\Rightarrow \{x_1, x_2\}$ is a linearly independent set

Extension of this claim: If $\lambda_1, \lambda_2, \dots, \lambda_n$ are distinct, then their eigenvectors $\{x_1, \dots, x_n\}$ are linearly independent

Pr: H.W.

Example:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{bmatrix}$$

Characteristic equation: $\lambda^2 - 4\lambda - 5 = 0$

Roots: $\lambda_1 = 5, \lambda_2 = -1$

Question: Is A diagonalizable? Yes, because it has 2 distinct eigenvalues

Finding eigenvectors: ① $\lambda_1 = 5$

$$A - \lambda_1 I = \begin{bmatrix} -4 & 4 \\ 2 & -2 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 1 \end{bmatrix} \in N(A - \lambda_1 I)$$

$$x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

② $\lambda_2 = -1$

$$A - \lambda_2 I = \begin{bmatrix} 2 & 4 \\ 2 & 4 \end{bmatrix}$$

$$\begin{bmatrix} -2 \\ 1 \end{bmatrix} \in N(A - \lambda_2 I)$$

$$x_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$S = \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} \quad \Lambda = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$

Check: $S^{-1} A S = \Lambda$

$$\begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 5 & 0 \\ 0 & -1 \end{bmatrix}$$

Remarks:

① $S^{-1}AS = \Lambda$ or $A = S\Lambda S^{-1}$

Is S unique? **NO** why? If x_1 is an eigenvector, then so is cx_1 ,

So, the columns of S can be scaled to get S' such that $S'^{-1}AS' = \Lambda$

② $A = S\Lambda S^{-1}$

Suppose col. 1 of S is y

Col. 1 of $S\Lambda = \lambda_1 y$, Col. 1 of $AS = Ay$

We know $AS = S\Lambda$

$Ay = \lambda_1 y \Rightarrow \lambda_1$ is an eigenvalue & y is an eigenvector of A

③ **Powers of A :** Suppose λ is an eigenvalue and x is an eigenvector of A

$$A^2 x = A(Ax) = \lambda Ax = \lambda^2 x$$

Suppose $S^{-1}AS = \Lambda$ Question: Is $S^{-1}A^2S = \Lambda^2$? **Yes.**

$$(S^{-1}AS)(S^{-1}AS) = (\Lambda)(\Lambda) \\ S^{-1}A^2S = \Lambda^2$$

The argument above works for a general $k \geq 1$, i.e.,

$$S^{-1}A^kS = \Lambda^k$$

④ "Not all matrices are diagonalizable"

Example: $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

Check: A does not have 2 linearly independent eigenvectors.

Lecture 4 : Solving Fibonacci Sequence using diagonalization

Example: Fibonacci sequence

0, 1, 1, 2, 3, 5, ... $\{F_k\}$

$$F_{k+2} = F_{k+1} + F_k$$

Q: What is F_{100} ?

A linear algebra solution to this question uses diagonalization.

System of equations: $F_{k+2} = F_{k+1} + F_k$

$$F_{k+1} = F_k + F_{k-1}$$

$$u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

$$u_{k+1} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} u_k \quad (\Rightarrow) \quad \begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = \begin{bmatrix} F_{k+1} + F_k \\ F_{k+1} \end{bmatrix}$$

$$\text{Let } A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad u_0 \xrightarrow{A} u_1 \xrightarrow{A} u_2 \xrightarrow{A} u_3 \rightarrow \dots$$

$$u_k = A^k u_0 \text{ is a solution to } u_{k+1} = A u_k$$

In a general scenario, where A is $n \times n$, u_k is $n \times 1$

If A is diagonalizable, i.e., A has " n " independent eigenvectors, then

$$u_0 = c_1 x_1 + c_2 x_2 + \dots + c_n x_n \quad \text{since } \{x_1, \dots, x_n\} \text{ is a basis}$$

$$\begin{aligned} A u_0 &= c_1 A x_1 + \dots + c_n A x_n & \Bigg| & \quad A^2 u_0 = A(c_1 \lambda_1 x_1 + \dots + c_n \lambda_n x_n) \\ &= c_1 \lambda_1 x_1 + \dots + c_n \lambda_n x_n & \Bigg| & \quad = c_1 \lambda_1^2 x_1 + \dots + c_n \lambda_n^2 x_n \end{aligned}$$

$$u_k = A^k u_0 = c_1 \lambda_1^k x_1 + \dots + c_n \lambda_n^k x_n$$

Back to Fibonacci: to find u_k , we need to check if $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$ is

diagonalizable. Find its eigenvectors, if you.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad \text{characteristic equation: } \lambda^2 - \lambda - 1 = 0$$

$$\lambda_1 = \frac{1 + \sqrt{5}}{2}, \quad \lambda_2 = \frac{1 - \sqrt{5}}{2}$$

$$u_0 = \begin{bmatrix} F_1 \\ F_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = c_1 x_1 + c_2 x_2$$

Let's find the eigenvectors of A :

$$(A - \lambda I)x = \begin{bmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{bmatrix} x = 0$$

$$x_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}, \quad x_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}$$

$$(A - \lambda_1 I) \begin{pmatrix} \lambda_1 \\ 1 \end{pmatrix} = \begin{bmatrix} \lambda_1^2 - \lambda_1 - 1 \\ 0 \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ since } \lambda_1 \text{ is a root of } \lambda^2 - \lambda - 1 = 0$$

$$\text{Similarly, } (A - \lambda_2 I) \begin{pmatrix} \lambda_2 \\ 1 \end{pmatrix} = 0$$

So, x_1 and x_2 are the eigenvectors corresponding to λ_1 and λ_2

Writing u_0 as a linear combination of x_1, x_2 :

$$u_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = c_1 \begin{bmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{bmatrix} \Rightarrow c_1 = \frac{1}{\sqrt{5}}, c_2 = -\frac{1}{\sqrt{5}}$$

$$u_k = c_1 \lambda_1^k x_1 + c_2 \lambda_2^k x_2$$

$$\begin{pmatrix} F_{k+1} \\ F_k \end{pmatrix} = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^k \begin{pmatrix} \frac{1+\sqrt{5}}{2} \\ 1 \end{pmatrix} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^k \begin{pmatrix} \frac{1-\sqrt{5}}{2} \\ 1 \end{pmatrix}$$

$$F_k = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^k - \frac{1}{\sqrt{5}} \underbrace{\left(\frac{1-\sqrt{5}}{2} \right)^k}_{< 1}$$

as k increases $\left(\frac{1-\sqrt{5}}{2} \right)^k$ becomes negligible

A good enough approximation $F_k \approx \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^k$

$$(\text{or}) \quad F_{100} \approx \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{100} = \frac{1}{\sqrt{5}} (1.618)^{100}$$

Bottom line: Diagonalizability can be used to understand linear recurrence relations.

Lecture 5 : Orthogonally diagonalizable matrices

Main result:

A is a real symmetric $n \times n$ matrix. Then,

- ① Eigenvalues of A are real
- ② Eigenvectors corresponding to different eigenvalues are linearly independent
- ③ A is orthogonally diagonalizable

↓ this means

$\exists Q$ satisfying $Q^T Q = I$ such that $\Rightarrow Q^{-1} = Q^T$

$$A = Q \Lambda Q^T$$

$$= \begin{bmatrix} 1 & & & \\ & \ddots & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} -x_1^T \\ -x_2^T \\ \vdots \\ -x_n^T \end{bmatrix}$$

Remark: ① A is diagonalizable + orthogonal matrix for diagonalization $\Rightarrow$ orthogonally diagonalizable

$$A = Q \Lambda Q^{-1}$$

$$Q^T Q = I$$

$$A = Q \Lambda Q^T$$

- ② Any real matrix is not necessarily diagonalizable, but a real symmetric is diagonalizable.
e.g. $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ not diagonalizable

Example:

$$A = \begin{bmatrix} 1 & -2 \\ -2 & -2 \end{bmatrix}$$

Eigenvalues: $\lambda_1 = -3, \lambda_2 = 2$

Eigenvectors: $x_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, x_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$$v_1 = \frac{x_1}{\|x_1\|} = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}, v_2 = \frac{x_2}{\|x_2\|} = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} v_1 & v_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \quad \text{check: } Q^T Q = I$$

Then, $Q \Lambda Q^T = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1/\sqrt{5} & 2/\sqrt{5} \\ -2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -2 & -2 \end{bmatrix} = A$

Check this.

Tutorial 4.1 : Polynomial Regression

Polynomial Regression

Let us understand polynomial regression by fitting following data points to a second degree polynomial

x	y
1	3
2	1.5
3	2.5

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 \quad \text{--- (1)}$$

$$y = \theta_0 + \theta_1 x + \theta_2 x^2 \quad \text{--- (1)}$$

(x_1, y_1) , (x_2, y_2) and (x_3, y_3)

$$y_1 = \theta_0 + \theta_1 x_1 + \theta_2 x_1^2 \quad \text{--- (2)}$$

$$y_2 = \theta_0 + \theta_1 x_2 + \theta_2 x_2^2 \quad \text{--- (3)}$$

$$y_3 = \theta_0 + \theta_1 x_3 + \theta_2 x_3^2 \quad \text{--- (4)}$$

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

$$\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \end{bmatrix}^{-1} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}$$

$$x_1 = 1 \quad y_1 = 3$$

$$x_2 = 2 \quad y_2 = 1.5$$

$$x_3 = 3 \quad y_3 = 2.5$$

$$\begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{4} \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}^{-1} \begin{bmatrix} 3 \\ 1.5 \\ 2.5 \end{bmatrix}$$

$$\theta_0 = 7$$

$$\theta_1 = -5.25$$

$$\theta_2 = 1.25$$

$$y = 7 + (-5.25)x + 1.25x^2$$

$$y = 7 - 5.25x + 1.25x^2$$

Tutorial 4.2 : Eigenvalues and Eigenvectors

Eigenvalues and Eigenvectors

- "Eigen" is a German word literally meaning "own" or "characteristic".
 - Solving an Eigen problem is like finding characteristic of a matrix.
 - Let's learn geometric interpretation of eigen values and eigen vectors.
 - When we apply linear transformation on a set of vectors then most of them are knocked off from their original basis.
 - The vectors which hold their original basis when linear transformation is applied are called Eigenvectors.
 - The values by which Eigenvectors are scaled during linear transformation are called Eigenvalues.
 - We can say that applying linear transformation on an eigenvector is similar to just scaling the eigenvector by the eigen value.
-

$$A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$$

$$|A - \lambda I| = 0$$

$$A - \lambda I = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -\lambda & 1 \\ -2 & -3-\lambda \end{bmatrix}$$

$$|-\lambda \quad 1 \\ -2 \quad -3-\lambda| = 0$$

$$-\lambda(-3-\lambda) + 2 = 0$$

$$3\lambda + \lambda^2 + 2 = 0$$

$$\lambda^2 + 3\lambda + 2 = 0$$

$$\lambda = -1, -2$$

Eigen vector demo

$$\lambda = -1 \quad Au = \lambda u \quad \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = -1 \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} x_2 \\ -2x_1 - 3x_2 \end{bmatrix} + \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$x_1 + x_2 = 0$$

$$x_1 = k \quad x_2 = -k$$

$$u = \begin{bmatrix} -1 \\ k \end{bmatrix} = k \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$u = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\lambda = -2 \quad u = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

$$\begin{array}{l} Au = \begin{bmatrix} -1 \\ 1 \end{bmatrix} = -u \\ Av = \begin{bmatrix} 2 \\ 4 \end{bmatrix} = -2v \end{array}$$

The Equation

$$\begin{aligned} A \begin{bmatrix} 1 \\ 1 \end{bmatrix} &= \begin{bmatrix} -1 \\ 5 \end{bmatrix} \\ A \begin{bmatrix} 1 \\ 2 \end{bmatrix} &= \begin{bmatrix} 3 \\ 0 \end{bmatrix} \end{aligned}$$

$$A\bar{x} = \lambda\bar{x}$$

- A = Transformation matrix
- $\bar{x}$ = *Eigenvector*
- λ = Corresponding Eigenvalue

$$(A - \lambda I)\bar{x} = 0 \quad (2)$$

$$\det(A - \lambda I) = 0 \quad (3)$$

- A = Transformation matrix
- $\bar{x}$ = Eigen Vector
- λ = Corresponding Eigenvalue
- I = Identity matrix
- Eqn 2 is usually used to find eigenvectors, and, Eqn 3 is used to find eigenvalues of a matrix.



Week 4 Live Session 1

1. Nullspace correction (last session)

$$A \rightarrow m \times n$$

$$\mathcal{N}(A) = \{x : Ax = 0, x \in \mathbb{R}^n\}$$

When is $\mathcal{N}(A) = \{0\}$?

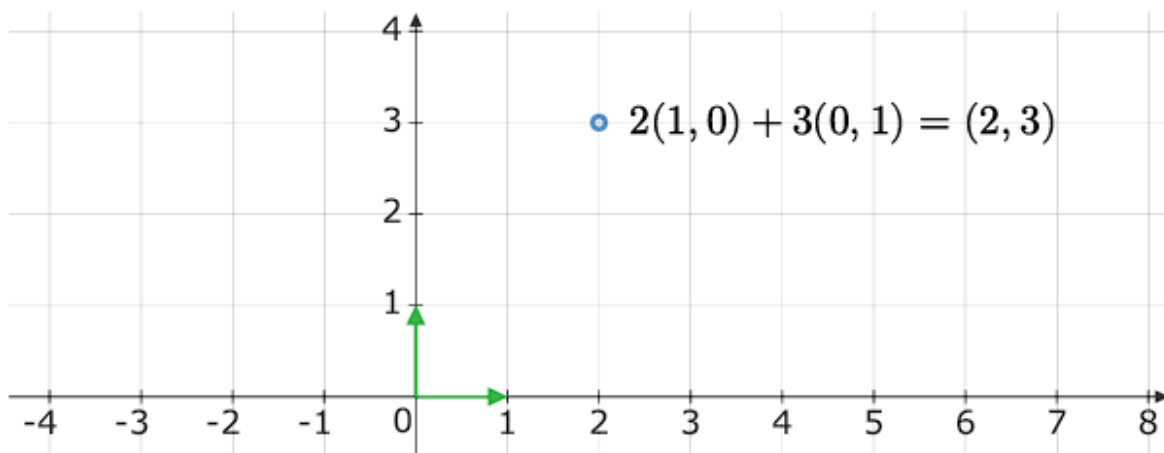
When the columns of A are linearly independent.

At 18:00 minutes in the previous session.

2. Linear Transformations

Every linear transformation (linear map) can be thought of as a matrix. In this case, we are interested in linear maps that go from $\mathbb{R}^n$ to $\mathbb{R}^n$.

Recall that a linear map can be exactly determined by what it does to the basis vectors in the domain. This is because every vector can be written as a unique linear combination of the basis vectors.

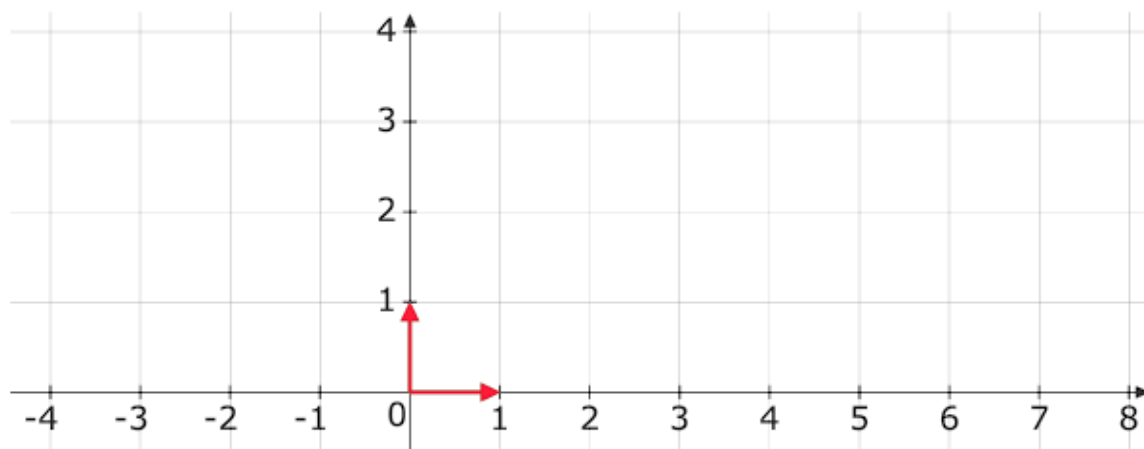


The columns of the matrix tell you where the basis vectors are going to go after the transformation.

2.1. Example-1

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

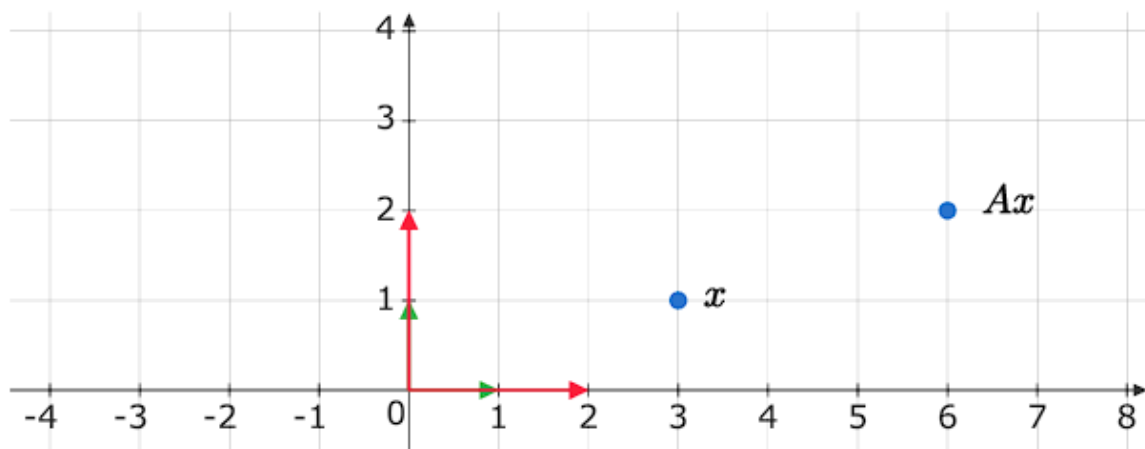
A is the identity map or the identity matrix.



2.2. Example-2

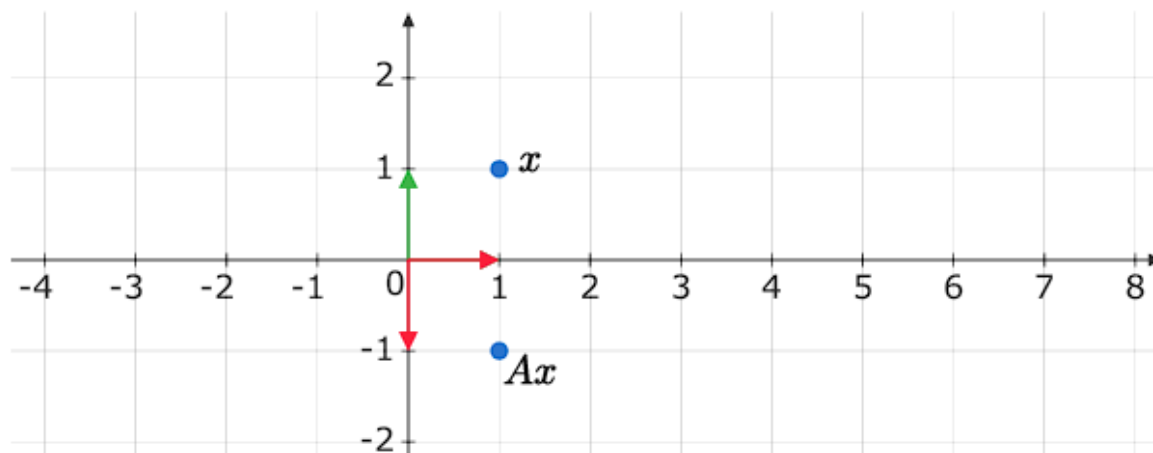
$$A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2I$$

A is scaling the basis vectors.



2.3. Example-3

$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

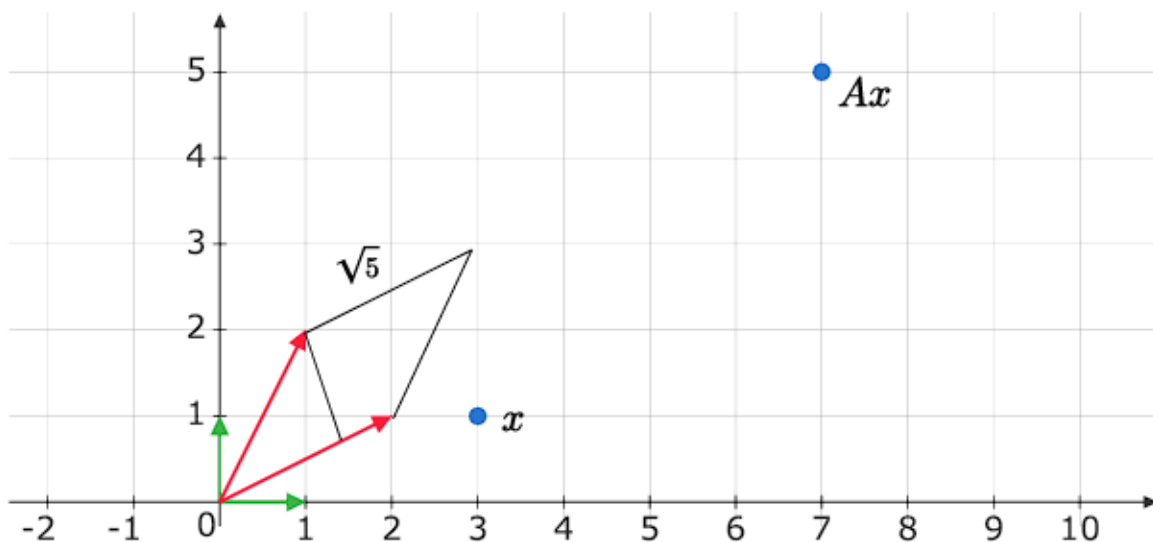


What is A doing to the basis vectors here? Rotation clockwise by 90° .

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

2.4. Example-4

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$



$$A \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 1 \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$$

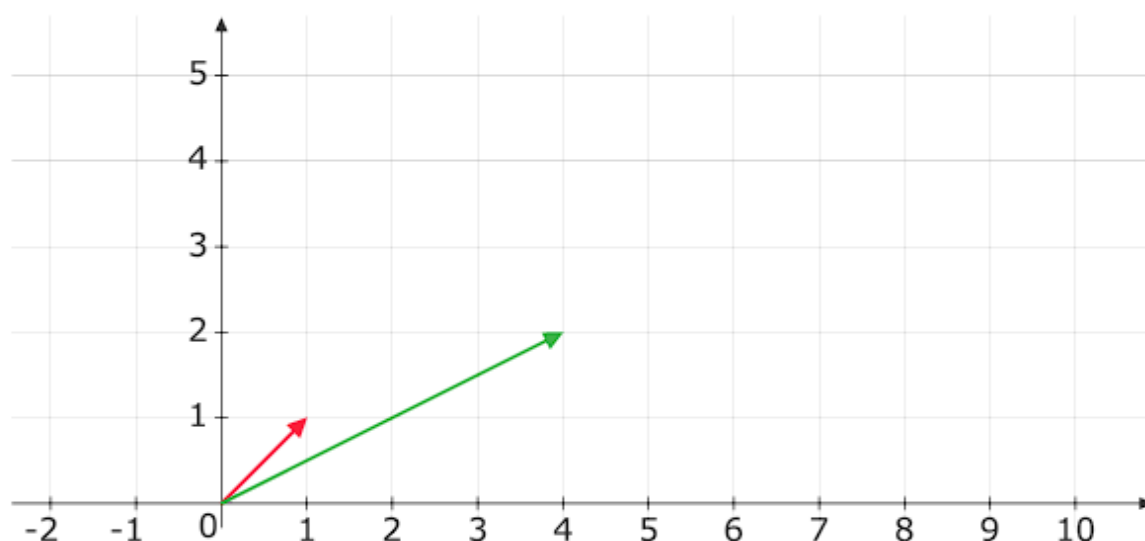
You can use this to understand determinants geometrically.

$$|A| = 3$$

The signed area of the parallelogram is actually equal to $|A|$. For a matrix that is not invertible, the area will turn out to be 0.

3. Eigenvalues and eigenvectors: Motivation

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$



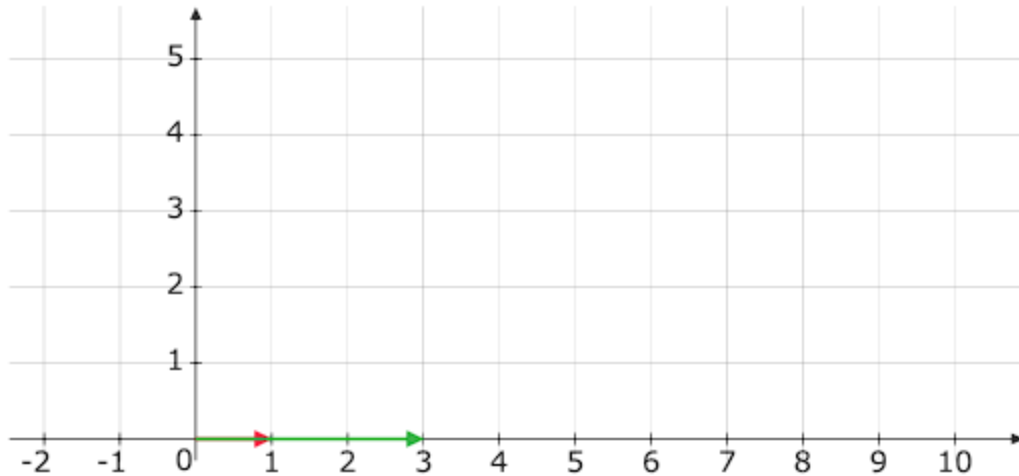
3.1. Case-1

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$$

(1, 1) and (4, 2) are pointing in different directions.

3.2. Case-2

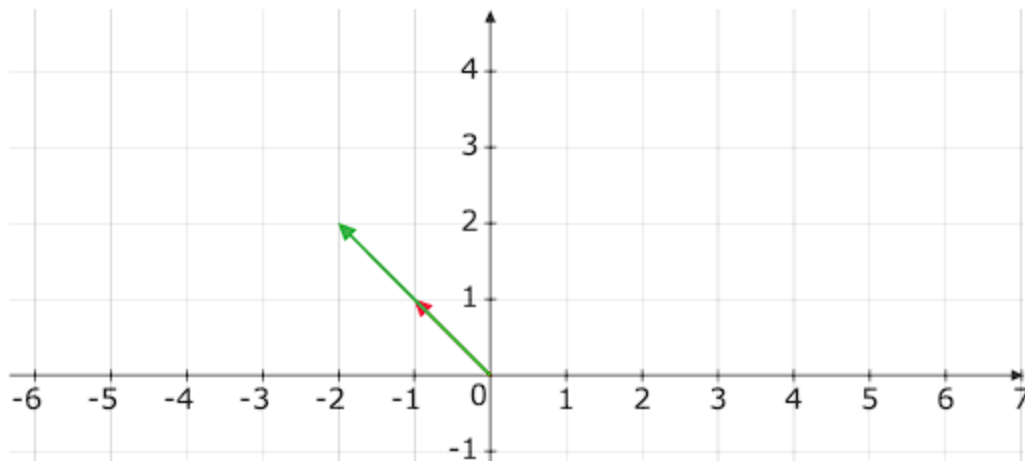
$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

(1,0) and (3,0) are in the same direction. A doesn't disturb the direction of the input vector.

$$A \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$



$$A \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 2 \end{bmatrix} = 2 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$(-1, 1)$ and $(-2, 2)$ are in the same direction.

Vectors for which the direction is preserved before and after the transformation are called eigenvectors. In this case, $(-1, 1)$ and $(1, 0)$ are eigenvectors of A . The extent by which the eigenvectors are scaled are called the eigenvalues. 2 and 3 are the corresponding eigenvalues.

Remark: The idea of the direction being preserved is a somewhat crude way of motivating the concept of eigenvalues and eigenvectors. Stick to the proper definition in the next section once you have got a rough idea of the concept.

4. Eigenvalues and eigenvectors: Definition

Formally, for a square matrix A of shape $n \times n$, a *non-zero* vector $v \in \mathbb{R}^n$ is called an eigenvector with eigenvalue λ if $Av = \lambda v$.

$$Av = \lambda v$$

For example,

- $(1, 0)$ is an eigenvector of $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ with eigenvalue 3.
- $(-1, 1)$ is an eigenvector of $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$ with eigenvalue 2.

Why should v be non-zero? What would happen if $v = 0$?

$$A0 = \lambda 0 \text{ for all } \lambda \in \mathbb{R}$$

We can't really pin down a particular value of λ and call that the eigenvalue.

What happens when $A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$?

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

Every vector (v_1, v_2) is an eigenvector with eigenvalue 0.

An eigenvector cannot have two distinct eigenvalues:

$$\begin{aligned} Av &= \lambda_1 v \\ Av &= \lambda_2 v \end{aligned} \implies \lambda_1 = \lambda_2$$

However, a single eigenvalue can be shared by multiple eigenvectors as we will see soon.

5. Eigenspace

If v is an eigenvector of A with eigenvalue λ , is v unique?

No, it is not unique. Let us consider the pair (λ, v) :

$$Av = \lambda v$$

Then:

$$\begin{aligned} A(kv) &= k(Av) \\ &= k(\lambda v) \\ &= \lambda(kv) \end{aligned}$$

Therefore, kv is an eigenvector with eigenvalue λ . In fact, if u and v are eigenvectors with the same eigenvalue λ .

$$\begin{aligned} A(u + v) &= Au + Av \\ &= \lambda u + \lambda v \\ &= \lambda(u + v) \end{aligned}$$

Therefore, $u + v$ is an eigenvector of A with eigenvalue λ .

Eigenspace, $E(\lambda)$

$$E(\lambda) = \{x : Ax = \lambda x, x \in \mathbb{R}^n\}$$

The eigenspace is actually a subspace of $\mathbb{R}^n$. We have shown the first two points in the preceding discussion.

- closed under addition
- closed under scalar multiplication
- $0 \in E(\lambda)$ because $A0 = \lambda 0$, however, 0 is not an eigenvector

$E(\lambda)$ contains the zero vector and the set of all eigenvectors of A corresponding to λ . Formally,

Eigenspace of an eigenvalue λ , denoted as $E(\lambda)$, is the set of all eigenvectors of A with eigenvalue λ along with the zero vector.

5.1. Example-1



$$A = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

0 is the only eigenvalue here, repeated twice.

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0 \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$E(0) = \mathbb{R}^2$$

5.2. Example-2

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

1 is an eigenvalue. In fact, 1 is the only eigenvalue of A , repeated twice.

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 1 \cdot \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

$$E(1) = \mathbb{R}^2$$

Can you have the following?

$$Au = \lambda_1 u$$

$$Av = \lambda_2 v$$

A matrix could have multiple eigenvalues.

6. Characteristic polynomial

How do we find the set of all eigenvalues for a matrix A ?

All the steps below are reversible.

x is an eigenvector
with eigenvalue λ

$$\iff Ax = \lambda x, \quad x \neq 0$$

$$\iff Ax = \lambda Ix, \quad x \neq 0$$

$$\iff Ax - \lambda Ix = 0, \quad x \neq 0$$

$$\iff (A - \lambda I)x = 0, \quad x \neq 0$$

$$\iff \mathcal{N}(A - \lambda I) \neq \{0\}$$

$$\iff A - \lambda I \text{ is not invertible}$$

$$\iff |A - \lambda I| = 0$$

x is an eigenvector with eigenvalue λ if and only if x is a solution to this homogeneous equation and $x \neq 0$. Therefore, $A - \lambda I$ has infinitely many solutions. This can happen only if $A - \lambda I$ is non-invertible, which in turn means that $|A - \lambda I| = 0$.

λ is an eigenvalue of A **if and only** if $|A - \lambda I| = 0$. Note that the eigenvectors do not make an appearance here.

We can temporarily forget about the eigenvectors and just worry about computing the eigenvalues.

$|A - \lambda I|$ is called the characteristic polynomial. This is a polynomial with degree n . The roots of the characteristic polynomial are the eigenvalues. Therefore, an $n \times n$ matrix can have at most n eigenvalues and also, the eigenvalues could repeat.

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

$$\begin{vmatrix} 3-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(2-\lambda) = 0$$

$$\implies \lambda = 2, 3$$

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & 3 & 3 \\ 3 & 1-\lambda & -1 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 0$$

$$(2-\lambda) \cdot [(1-\lambda)^2 - 9] = 0$$

$$(2-\lambda)(4-\lambda)(-2-\lambda) = 0$$

$$\lambda = 2, 4, -2$$

Lemma: Eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are orthogonal.

Proof: Assume that $\lambda_1 \neq \lambda_2$ are eigenvalues of a symmetric matrix A .

$$(\lambda_1, v_1), (\lambda_2, v_2)$$

Then:

$$Av_1 = \lambda_1 v_1$$

$$Av_2 = \lambda_2 v_2$$

Now:

$$v_2^T Av_1 = \lambda_1 v_2^T v_1$$

$$v_2^T A^T v_1 = \lambda_1 v_2^T v_1$$

$$(Av_2)^T v_1 = \lambda_1 v_2^T v_1$$

$$\lambda_2 v_2^T v_1 = \lambda_1 v_2^T v_1$$

$$(\lambda_2 - \lambda_1) v_2^T v_1 = 0$$

Therefore, $v_2^T v_1 = 0$.

7. Characteristic Polynomials: Some Observations

- The characteristic polynomial has degree n .
- There are at most n eigenvalues.
- The sum of the eigenvalues is equal to the trace of the matrix.
- The product of the eigenvalues is equal to the determinant of the matrix.

$$|A - \lambda I| = (\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$$

If you set $\lambda = 0$ then $|A| = \lambda_1 \cdots \lambda_n$.

- There are matrices that have no eigenvalues.

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{vmatrix} 1-\lambda & -1 \\ 1 & 1-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)^2 = -1$$

A doesn't have any real eigenvalues.

Remark: For weeks 3 and 4, we are only concerned with real vector spaces.

Can a matrix have eigenvectors without having any eigenvalues?

No. This is absurd. Eigenvectors and eigenvalues are inseparable.

8. Computing Eigenvectors (Eigenspace)

$$A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$$

A has eigenvalues 3, 2. Let us now compute the eigenvectors for $\lambda = 2$.

$$(A - 2I)v = 0$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

v_2 is a free variable. Setting $v_2 = 1$, $v_1 = -1$. Therefore, $(-1, 1)$ is an eigenvector. In fact, the eigenspace $E(2)$:

$$E(2) = \text{span}\{(-1, 1)\}$$

You have to essentially solve the system of homogeneous equations.
Alternatively, you can look at it as first finding a basis for the nullspace of

$$A - \lambda I.$$

$$(A - 3I)v = 0$$

$$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & \mathbf{1} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

The second column contains the pivot. So the second variable is the pivot variable. v_1 is the free variable. So setting $v_1 = 1$, we get $v_2 = 0$. Therefore, $(1, 0)$ is an eigenvector.

$$E(3) = \text{span}\{(1, 0)\}$$

$$A = \begin{bmatrix} 1 & 3 & 3 \\ 3 & 1 & -1 \\ 0 & 0 & 2 \end{bmatrix}$$

We know that $\lambda = 2, 4, -2$.

$$(A + 2I)v = 0$$

$$\begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & -1 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 3 & 3 \\ 3 & 3 & -1 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow{R_2 \rightarrow R_2 - R_1} \begin{bmatrix} 3 & 3 & 3 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix}$$

$$\begin{aligned}
 & \begin{bmatrix} 3 & 3 & 3 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow{R_1 \rightarrow \frac{R_1}{3}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix} \\
 & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 4 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{bmatrix} \\
 & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_2 \rightarrow \frac{R_2}{-4}} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \\
 & \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - R_2} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}
 \end{aligned}$$

$$\begin{bmatrix} \mathbf{1} & 1 & 0 \\ 0 & 0 & \mathbf{1} \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- v_1 and v_3 are pivot (dependent) variables
- v_2 is the only free variable

$$v_1 = -1$$

$$v_2 = 1$$

$$v_3 = 0$$

Therefore, $(-1, 1, 0)$.

$$E(-2) = \text{span}\{(-1, 1, 0)\}$$

So far all the examples that we have looked have an eigenspace that is one-dimensional. Is this always the case?

No. For example, if $A = I$, the eigenspace is $\mathbb{R}^n$.

9. Eigenthings of Special Matrices

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

The characteristic polynomial of D is:

$$|D - \lambda I| = 0$$

$$\begin{vmatrix} 1 - \lambda & 0 & 0 \\ 0 & 2 - \lambda & 0 \\ 0 & 0 & 3 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(2 - \lambda)(3 - \lambda) = 0$$

The eigenvalues of a diagonal matrix are its diagonal elements. One can directly read the eigenvalues off the diagonal matrix with just a lookup. $D[i][i]$. This is $O(1)$, computationally efficient.

What are the eigenvectors of D ? The eigenvectors are the elements of the standard basis.

$$D = \begin{bmatrix} 1 & & \\ & 1 & \\ & & 0 \end{bmatrix}$$

$E(1)$ is a two-dimensional subspace of $\mathbb{R}^3$.

$$T = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

T is upper-triangular.

$$|T - \lambda I| = 0$$

$$\begin{vmatrix} 1 - \lambda & 3 & 1 \\ 0 & 2 - \lambda & 2 \\ 0 & 0 & 3 - \lambda \end{vmatrix} = 0$$

$$(3 - \lambda)(1 - \lambda)(2 - \lambda) = 0$$

The eigenvalues of an upper-triangular (or lower triangular) matrix are the diagonal elements.

Projection matrices:

$$P^2 = P \text{ and } P^T = P$$

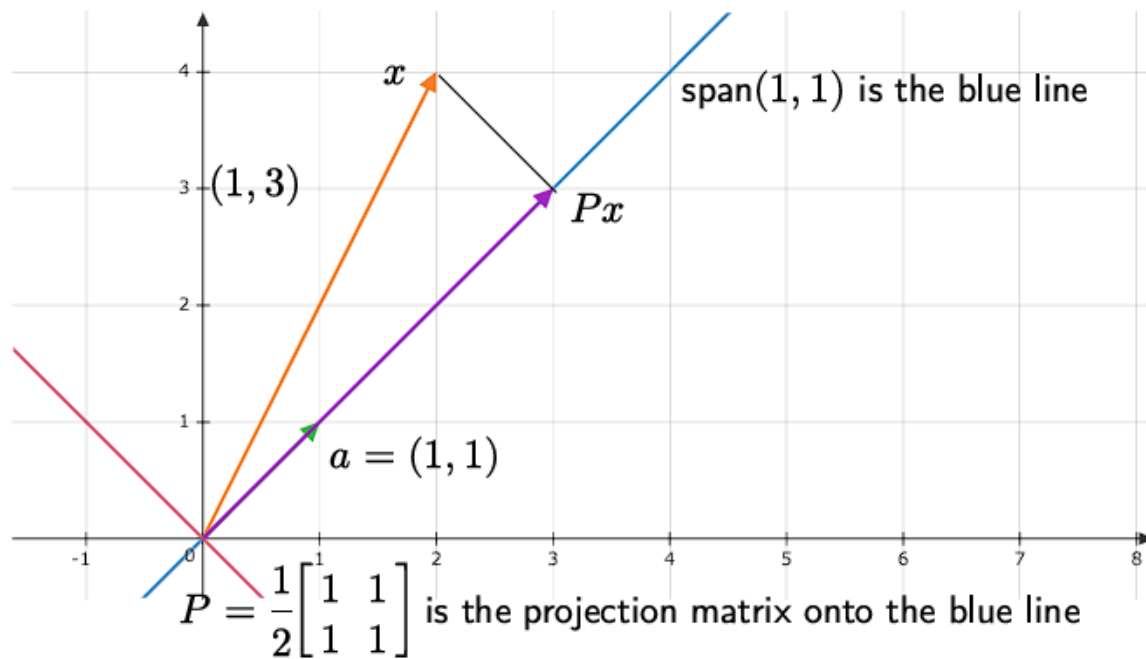
The eigenvalues are 0 and 1. If you pick an element v from the column space of P , what is $Pv = v$. 1 is an eigenvalue and every element in the column space of P is an eigenvector.

$$aa^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Outer products characterize rank-1 matrices. Any matrix can be decomposed as a sum of rank-1 matrices (SVD). This can help you break down a matrix as a sum of smaller matrices.

- 50×5 matrix
- 50 entries in an excel sheet with 5 features describing each entry

Low rank approximation. LoRA in LLMs.



$$a^T a = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2$$

$$P = \frac{aa^T}{a^T a} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

$$\begin{aligned} P^2 &= \frac{1}{4} \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \\ &= P \end{aligned}$$

Verify. If $x = (2, 4)$ then Px is the projection of x onto the line passing through $(1, 1)$.

$$P \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$P \begin{bmatrix} k \\ k \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} k \\ k \end{bmatrix} = 1 \cdot \begin{bmatrix} k \\ k \end{bmatrix}$$

Notice that (k, k) is an eigenvector of P with eigenvalue 1.

$$P \begin{bmatrix} k \\ -k \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} k \\ -k \end{bmatrix} = 0 \cdot \begin{bmatrix} k \\ -k \end{bmatrix}$$

Notice that $(k, -k)$ is an eigenvector of P with eigenvalue 0.

Week 4 Live Session 2

1. Recap

- Eigenvalues and eigenvectors

$$Ax = \lambda x$$

- Characteristic polynomial

$$|A - \lambda I| = 0$$

- Computing eigenvectors

$$(A - \lambda I)v = 0$$

2. Diagonal matrices

A diagonal matrix is given by $D = \text{diag}(a_1, \dots, a_n)$.

$$D = \begin{bmatrix} a_1 & & \\ & \ddots & \\ & & a_n \end{bmatrix}$$

Determinant

$$|D| = a_1 \cdots a_n$$

Invertibility and Rank

A diagonal matrix is invertible if:

- $|D| \neq 0$
- None of the diagonal entries are zero
- All diagonal entries are non-zero

If D is invertible, then:

$$D^{-1} = \text{diag}\left(\frac{1}{a_1}, \dots, \frac{1}{a_n}\right)$$

The rank of a diagonal matrix is the number of non-zero diagonal entries.

Eigenvalues

The characteristic polynomial is just:

$$|D - \lambda I| = (a_1 - \lambda) \cdots (a_n - \lambda)$$

Therefore, the eigenvalues of a diagonal matrix are the entries on the main diagonal.

Eigenvectors

If a_i is the i^{th} diagonal entry of D , then e_i is an eigenvector of D corresponding to eigenvalue a_i , where $e_i = (0, \dots, 1, \dots, 0)$ is the i^{th} element in the standard basis. For example:

$$\begin{aligned} De_2 &= \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ a_2 \\ 0 \end{bmatrix} \\ &= a_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \end{aligned}$$

3. Diagonalizability: Definition-1



Diagonal matrices have very useful properties. Can we find matrices that are **similar** to diagonal matrices? Can we find matrices that share some of these properties of diagonal matrices, at least indirectly?

Definition-1: A matrix A of shape $n \times n$ is called **diagonalizable** if it is similar to a diagonal matrix of the same shape.

Remark-1: Two matrices A and B of shape $n \times n$ are similar if there is some invertible matrix P of the same shape such that:

$$B = PAP^{-1}$$

Remark-2: Similar matrices actually represent the same linear transformation in different bases. One can view the invertible matrix P as a change-of-basis matrix. This is what is called a similarity transformation. For example, consider:

$$T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x + y, x - y)$$

Basis: $\beta = \{(1, 0), (0, 1)\}$

$$A = [T]_{\beta}^{\beta} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Basis: $\gamma = \{(1, 1), (-1, 1)\}$

We now need to express $T(1, 1) = (2, 0)$ and $T(-1, 1) = (0, -2)$ in the new basis γ . We see that $(2, 0) = (1, 1) - (-1, 1)$ and $(0, -2) = -(1, 1) - (-1, 1)$. Therefore:

$$B = [T]_{\gamma}^{\gamma} = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}$$

A and B are similar. Meaning, there is some invertible matrix that connects A and B . The columns of this invertible matrix is actually the elements of the basis γ :

$$P = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

Now, we can verify that:

$$A = PBP^{-1}$$

Diagonalizable matrices:

$$A = PDP^{-1}$$

We will study the consequences of this definition. If we right-multiply by P on both sides:

$$AP = PD$$

$$P = \begin{bmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{bmatrix}, \quad D = \begin{bmatrix} \lambda_1 & & \\ & \lambda_2 & \\ & & \lambda_3 \end{bmatrix}$$

$$AP = \begin{bmatrix} | & | & | \\ Av_1 & Av_2 & Av_3 \\ | & | & | \end{bmatrix}$$

$$PD = \begin{bmatrix} | & | & | \\ \lambda_1 v_1 & \lambda_2 v_2 & \lambda_3 v_3 \\ | & | & | \end{bmatrix}$$

$$Av_i = \lambda_i v_i$$

This means that v_i is an eigenvector of A with eigenvalue λ_i . Can v_i be zero? It cannot be zero as v_i is a column of an invertible matrix. This means that we are justified in calling v_i an eigenvector.

4. Diagonalizability: Definition-2

- P is an invertible matrix.
- The columns of P are eigenvectors of A .
- Therefore, there is a basis for $\mathbb{R}^n$ made up of eigenvectors of A .

Definition-2: A matrix A of shape $n \times n$ is diagonalizable if there is a basis for $\mathbb{R}^n$ made up of eigenvectors of A .

5. Eigen-decomposition

A diagonalizable matrix can be broken down into the product of three matrices which contain information about the eigenvalues and eigenvectors of the matrix:

$$A = PDP^{-1}$$

How do you diagonalize a matrix that is diagonalizable?

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

Step-1: Find the eigenvalues

$$\begin{aligned} |A - \lambda I| &= (1 - \lambda)(4 - \lambda) - 10 \\ &= \lambda^2 - 5\lambda - 6 \\ &= (\lambda - 6)(\lambda + 1) \end{aligned}$$

Therefore, the eigenvalues are $\lambda = 6, -1$. Always a good practice (convention) to list the eigenvalues in decreasing order.

Step-2: Find the eigenvectors

$$(A - 6I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -5 & 2 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-5v_1 + 2v_2 = 0$$

$$\begin{aligned}v_1 &= 2 \\v_2 &= 5\end{aligned}$$

Therefore, $(2, 5)$ is an eigenvector for A corresponding to $\lambda = 6$

Verify:

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} \rightarrow 2(1, 5) + 5(2, 4) = (12, 30) = 6(2, 5)$$

$$E(6) = \text{span}\{(2, 5)\}$$

$$\begin{aligned}(A + I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \begin{bmatrix} 2 & 2 \\ 5 & 5 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ 2v_1 + 2v_2 &= 0\end{aligned}$$

$(1, -1)$ is an eigenvector of A corresponding to $\lambda = -1$.

Verify:

$$(1, 5) - (2, 4) = (-1, 1) = -1(1, -1)$$

Step-3: Collect the eigenvalues in a diagonal matrix and the corresponding eigenvectors go into the columns of P

$$D = \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}, P = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix}$$

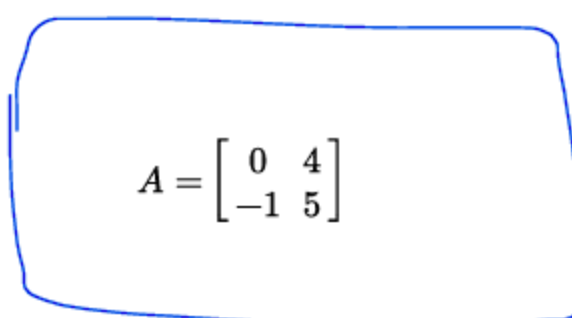
Step-4: Find P^{-1} and express $A = PDP^{-1}$

$$P^{-1} = \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix}$$

Verify:

$$\begin{aligned}PDP^{-1} &= \frac{1}{-7} \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} \\&= \frac{-1}{7} \begin{bmatrix} 12 & -1 \\ 30 & 1 \end{bmatrix} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix} \\&= \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix} \\&= A\end{aligned}$$


Homework


$$A = \begin{bmatrix} 0 & 4 \\ -1 & 5 \end{bmatrix}$$

6. Properties

In all these properties, A is some diagonalizable matrix:

$$A = PDP^{-1}$$

Matrix power

$$\begin{aligned}A^2 &= (PDP^{-1})(PDP^{-1}) \\&= PDDP^{-1} \\&= PD^2P^{-1}\end{aligned}$$

In general:

$$A^k = PD^kP^{-1}$$

Eigenvalues and eigenvectors

$$\begin{aligned}|A - \lambda I| &= |PDP^{-1} - \lambda I| \\&= |PDP^{-1} - \lambda PIP^{-1}| \\&= |P(DP^{-1} - \lambda IP^{-1})| \\&= |P(D - \lambda I)P^{-1}| \\&= |P| \cdot |D - \lambda I| \cdot |P^{-1}| \\&= |D - \lambda I|\end{aligned}$$

A and D have the same characteristic polynomial and hence the same eigenvalues. Moreover, A has a full set of n eigenvalues.

Important: If an $n \times n$ matrix is diagonalizable, it has n eigenvalues, not necessarily distinct. When we say that there are n eigenvalues, we also include the multiplicity of eigenvalues. For example, for the identity matrix I of shape $n \times n$, there are n eigenvalues, but only one unique eigenvalue, namely 1, that is repeated n times.

True: If A is diagonalizable then A has n eigenvalues.

Invertibility

If A is invertible, then:

$$A^{-1} = PD^{-1}P^{-1}$$

Determinant

$$|A| = |PDP^{-1}| = |D|$$

Eigenvectors

Are the eigenvectors of A and D equal? Not necessarily.

- The eigenvectors of A can be read off by looking at the matrix P .
The columns of P are the eigenvectors of A .
- The eigenvectors of D are the elements of the standard basis of $\mathbb{R}^n$.

7. Tests for Diagonalizability

How do we know if a given matrix is diagonalizable?

7.1. Negative Test

Consider the matrix A :

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$|A - \lambda I| = (1 - \lambda)^2 + 1$$

What are the eigenvalues? A doesn't have any real eigenvalues. It is alright to say that A doesn't have any eigenvalues while dealing with real vector spaces.

Remark: For weeks 3 and 4, we are only dealing with real vector spaces. Therefore, eigenvalues and eigenvectors, if they exist, have to be real.

Can A be diagonalizable? No. This is because if A had been diagonalizable, it would have had 2 eigenvalues.

Negative Test: If a matrix A of shape $n \times n$ doesn't have n eigenvalues (including repetitions) then A is not diagonalizable.

7.2. Positive Test

Consider a matrix A that has two distinct eigenvalues, say λ_1, λ_2 , where $\lambda_1 \neq \lambda_2$.

Theorem: Eigenvectors corresponding to distinct eigenvalues are linearly independent.

Let v_1, v_2 be eigenvectors corresponding to λ_1, λ_2 :

$$\alpha_1 v_1 + \alpha_2 v_2 = 0$$

$$(A - \lambda_1 I)(\alpha_1 v_1 + \alpha_2 v_2) = 0$$

$$\alpha_1(\lambda_1 - \lambda_1)v_1 + \alpha_2(\lambda_2 - \lambda_1)v_2 = 0$$

$$\alpha_2(\lambda_2 - \lambda_1)v_2 = 0$$

$\alpha_2 = 0$ because $\lambda_1 \neq \lambda_2$ and $v_2 \neq 0$ (v_2 is an eigenvector). If $\alpha_2 = 0$ then $\alpha_1 = 0$. Therefore, $\{v_1, v_2\}$ is linearly independent.

Positive Test: If a matrix A of shape $n \times n$ has n distinct eigenvalues, then A is diagonalizable.

Why?

- If there are n distinct eigenvalues, then the corresponding set of n eigenvectors are linearly independent.
- We now have a basis of eigenvectors for $\mathbb{R}^n$ and hence A is diagonalizable.

7.3. Inconclusive?

What if a matrix A of shape $n \times n$ has n eigenvalues where some of them repeat? Is such a matrix diagonalizable?

This is an inconclusive test. You can't comment on the diagonalizability of A .

8. Diagonalizable Matrices: Examples

Homework

1

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$$

2

$$A = \begin{bmatrix} 2 & 0 & -3 \\ 1 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

Step-1

$$\begin{aligned} |A - \lambda I| &= (-1 - \lambda)(2 - \lambda)(-1 - \lambda) \\ &= (2 - \lambda)(1 + \lambda)^2 \end{aligned}$$

Eigenvalues are 2, -1, -1.

Step-2:

$$(A - 2I) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 0 & 0 & -3 \\ 1 & -3 & -1 \\ 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & -3 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

v_1 and v_3 are pivot variables and v_2 is a free variable.

$$\left. \begin{array}{l} v_1 = 3 \\ v_2 = 1 \\ v_3 = 0 \end{array} \right\}$$

$(3, 1, 0)$ is an eigenvector corresponding to $\lambda = 2$.

$$(A + I) \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 3 & 0 & -3 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = 0$$

$\lambda = -1$ will have two linearly independent eigenvectors.

A set of linearly independent eigenvectors for $\lambda = -1$:

$$\{(1, 0, 1), (1, 1, 1)\}$$

Therefore, a basis for $\mathbb{R}^3$ made up of eigenvectors of A :

$$\beta = \{(3, 1, 0), (1, 0, 1), (1, 1, 1)\}$$

Therefore A is diagonalizable.

$$P = \begin{bmatrix} 3 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 2 & & \\ & -1 & \\ & & -1 \end{bmatrix}$$

Verify:

$$A = PDP^{-1}$$

This is an eigen-decomposition of A .

9. Non-Diagonalizable Matrices



9.1. Example-1

$$A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$$



Step-1: Find eigenvalues

$$\begin{aligned} |A - \lambda I| &= (3 - \lambda)(1 - \lambda) + 1 \\ &= \lambda^2 - 4\lambda + 4 \\ &= (\lambda - 2)^2 \end{aligned}$$



Step-2: Find the eigenvectors

$$(A - 2I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



$(1, 1)$ is an eigenvector of A with eigenvalue $\lambda = 2$.



$$E(2) = \text{span}\{(1, 1)\}$$

- Can you find a basis of eigenvectors for $\mathbb{R}^2$?
- Can you find an invertible matrix whose columns are the eigenvectors of A ?

No

No

No, we can't.

$$P = \begin{bmatrix} 2 & 5 \\ 2 & 5 \end{bmatrix} \rightarrow \text{rank}(P) = 1$$

Conclusion is that $A = \begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$ is not diagonalizable.

9.2. Example-2

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$$

Step-1:

$$\begin{aligned} (2 - \lambda)^2 &= 0 \\ \Rightarrow \lambda &= 2 \end{aligned}$$

Step-2:

$$\begin{aligned} (A - 2I) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= 0 \\ \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} &= 0 \Rightarrow v_1 = 1, v_2 = 0 \\ v_2 &= 0 \end{aligned}$$

$(1, 0)$ is an eigenvector of A with eigenvalue $\lambda = 2$. You don't have a set of two independent eigenvectors for A , hence A is not diagonalizable.
There is no basis for the underlying linear transformation in which the matrix representation is diagonal.

9.3. Example-3

Homework

$$A = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & -1 \\ 2 & 0 & 2 \end{bmatrix}$$



10. Problems

10.1. Problem-1

Find A^k where:

$$A = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

Refer to section-5 for the eigen-decomposition of A :

$$D = \begin{bmatrix} 6 & 0 \\ 0 & -1 \end{bmatrix}, P = \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix}, P^{-1} = \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix}$$

$$A = PDP^{-1}$$

$$A^k = PD^kP^{-1}$$

$$= \begin{bmatrix} 2 & 1 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} 6^k & 0 \\ 0 & (-1)^k \end{bmatrix} \frac{1}{-7} \begin{bmatrix} -1 & -1 \\ -5 & 2 \end{bmatrix}$$

Relate this to Fibonacci sequence example:

$$\begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$



Refer to this [document](#) for the details of the Fibonacci example discussed in the lectures.

11. Orthogonally Diagonalizable Matrices

Orthogonal matrices are square matrices Q that satisfy $Q^T Q = Q Q^T = I$. Therefore, $Q^{-1} = Q^T$. The columns of Q are orthonormal, they in fact form an orthonormal basis for $\mathbb{R}^n$.

Real Spectral Theorem: All symmetric matrices are diagonalizable. In fact, they are orthogonally diagonalizable. It means that if A is symmetric, then there is an orthogonal matrix Q and a diagonal matrix D such that:

$$A = Q D Q^T$$

This is often called the spectral decomposition.

Lemma: Eigenvectors corresponding to distinct eigenvalues of a symmetric matrix are orthogonal.

For week-5, not for quiz-1

Conjugate transpose: generalization of transpose to complex matrices:

$$A = \begin{bmatrix} 1+i & i \\ 2-i & 3i \end{bmatrix} \Rightarrow A^* = \begin{bmatrix} 1-i & 2+i \\ -i & -3i \end{bmatrix}$$

$$A^* = (\overline{A})^T$$

Hermitian matrix is one which is equal to its conjugate transpose matrix:

$$A^* = A$$

$$A = \begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix}$$

$$A^* = \begin{bmatrix} 1 & 1+i \\ 1-i & 2 \end{bmatrix} = A$$

11.1. Problem-2

For what values of k is $A = \begin{bmatrix} 5 & 3 \\ k & -1 \end{bmatrix}$ diagonalizable over $\mathbb{R}$?

Answer: A is diagonalizable whenever $k > -3$.

Step-1: Find the characteristic polynomial

$$\begin{aligned} & (5 - \lambda)(-1 - \lambda) - 3k \\ &= \lambda^2 - 4\lambda - 5 - 3k \\ &= \lambda^2 - 4\lambda + (-5 - 3k) \end{aligned}$$

Step-2: Ensure real eigenvalues

$$\Delta = 16 + 4(5 + 3k)$$

For real eigenvalues, $\Delta \geq 0$:

$$\begin{aligned}\Delta &\geq 0 \\ 16 + 4(5 + 3k) &\geq 0 \\ 4 + 5 + 3k &\geq 0 \\ k &\geq -3\end{aligned}$$

If $k \geq -3$, A has two real eigenvalues. If the discriminant of a quadratic equation is greater than or equal to zero, then it has two real roots:

$$\lambda = \frac{4 \pm \sqrt{\Delta}}{2}$$

Step-3:

If $k > -3$, what can you say? A has two distinct eigenvalues, which means that A is diagonalizable.

Step-4:

Look at what happens when $k = -3$, repeated roots, or one eigenvalue repeated twice. Find this eigenvalue:

$$\lambda = 2, 2$$

$$(A - 2I)v = 0$$

$$\begin{bmatrix} 3 & 3 \\ -3 & -3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$(1, -1)$ is an eigenvector for A with eigenvalue for $\lambda = 2$. Since we don't have two linearly independent eigenvectors for A , A is not diagonalizable.

PA : 1,9,10,14,15

GA : 11,12,14,15,16